Bipin Shrestha

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SUMMARY

Motivated Computer Science student passionate about software development and problem-solving. Proficient in Java with strong algorithms and data structures skills, experienced in full-stack web development and cloud/DevOps projects, AWS Certified Solutions Architect, eager to contribute innovative solutions and grow in a dynamic tech environment.

EDUCATION

University of Louisiana at Monroe — Monroe, LA

Bachelor of Science in Computer Science

Data Structures and Algorithms, Object-Oriented Programming, Operating System, Computer Networks, Database Management

PROJECTS

Grade Forge – Automated Programming Assignment Grading Platform *Website | Repository*

A scalable platform for managing programming assignments where instructors can create test suites and automatically evaluate student code submissions using containerized execution.

- Led the backend development of a **team-based project**, designing core system architecture and coordinating implementation of the grading pipeline and API structure.
- Built a secure platform where **Faculty create assignments and test suites**, and students submit code that is compiled and executed inside **isolated Docker containers** for automated grading.
- Implemented a **test case execution engine** that evaluates submissions against multiple test cases, storing compile results, runtime output, execution time, and pass/fail status for each submission.
- Designed **role-based access control** (Faculty, TA, Student) using Spring Security to manage assignment creation, grading workflows, and submission tracking.
- Deployed the system on AWS EC2 with **AWS RDS (PostgreSQL)** for database management and **AWS S3** for storing submission files and artifacts.
- Designed the system to support **future enhancements** such as distributed code runners, asynchronous grading queues, plagiarism detection, and grading analytics dashboards.

Tech Stack: Java, Spring Boot, React, TailwindCSS, Docker, PostgreSQL, AWS EC2, AWS RDS, AWS S3, AWS CodeBuild, AWS IAM, Spring Security, REST APIs, GitHub, Hibernate

Hotel Booking Application *Website | Repository*

A full-stack hotel booking system for privately-owned or large hotels, built with Spring Boot, Next.js, and AWS.

- Implemented secure JWT-based authentication with role-based access for users, supporting sign-up, login, bookings, payments, and profile management.
- Developed admin dashboard for room management (CRUD) and booking approvals, integrated with asynchronous email notifications and Stripe payments.
- Deployed backend on AWS (EC2, RDS, S3) with CI/CD, reducing **deployment time from 30 mins to <5 mins**.

Tech Stack: *Java, Spring Boot, Spring, Next.js, AWS, MySQL, Tailwind CSS, JavaScript*

Research & Collaboration

Emerging Scholar Program Research *Spring 2023 – Fall 2023*

Academic Research -University of Louisiana at Monroe

- VR in Education – analyzed teacher perceptions of virtual reality for enhanced student engagement and experiential learning
- AI in Business Communication – Studied impacts of AI, Robotics, and ICT on modern business operations, workforce Management, and global economic trends.

Google Developer Code (GDSC) *Fall 2024 - Present*

Active Community Member – Coding competition, hackathons, and mentoring fellow developers.

Technical Skill

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5/CSS3, Tailwind CSS, Chart.js, Responsive UI

Backend & APIs: Java, Spring Boot, Spring AI, REST APIs, Git, JWT Authentication & Role-Based Access, AOP, Microservices

Databases & Data: MySQL, PostgreSQL, MongoDB, Hibernate/JPA ORM, Data Modeling, Caching

AI & Automation: OpenRouter API, Generative AI, Spring AI, n8n, MCP

DevOps & Cloud / Project Management: Docker, RabbitMQ, AWS (EC2, RDS, S3, API Gateway, Elastic Beanstalk, SNS, SQS, DynamoDB), CI/CD (CodePipeline & CodeBuild), Linux, GitHub, Jira